

Order Flow Imbalance and Price Impact in CME Ether Futures

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Abstract

We study Order Flow Imbalance (Cont, Kukanov, and Stoikov, 2014) on CME Ether futures from February 2021 to August 2026. On 28,711 out-of-sample dollar-volume bars, an OLS regression of bar mid-quote change on contemporaneous and one-bar lagged OFI with Newey-West HAC standard errors yields a contemporaneous coefficient $\hat{\beta}_0 = +0.215$ ($t = 58.1$, $R^2 = 0.30$), against $+0.300$ and $R^2 = 0.31$ in sample. The fit does not depend on the bar construction: rebuilt at scales from ten-second intervals to bars spanning hours, the out-of-sample R^2 stays between 0.29 and 0.32. The relation is concave, with impact growing roughly as the 0.58 power of flow; it is symmetric in sign and steeper when the quoted spread is wide. The regression is contemporaneous, so it measures price impact rather than forecasting power: used on its own as a predictor, lagged OFI explains essentially none of the following bar’s move ($R^2 < 0.001$ out of sample). OFI is a strong descriptor of price formation on this contract; it is not, by itself, a trading signal.

1 Introduction

Order Flow Imbalance (OFI) aggregates best-bid and best-ask quote updates into a signed measure of top-of-book pressure. On liquid US equities, contemporaneous OFI explains roughly two-thirds of mid-quote variation over ten-second intervals (Cont, Kukanov, and Stoikov, 2014). The cryptocurrency evidence is thinner: Silantsev (2019) finds trade-flow imbalance outperforms OFI in explaining short-horizon price changes on BitMEX XBT/USD, and Bieganowski and Ślepaczuk (2026) report OFI as the top SHAP feature in CatBoost ensembles on Binance Futures perpetuals. CME’s Ether contract sits apart from both: it trades on a regulated venue, and its quoted spread at trade instants has a panel median near four basis points of price.

This paper estimates the contemporaneous price impact of OFI on CME Ether futures from February 2021 to August 2026. The construction is the level-1 (best bid / best ask) form of Cont et al. on dollar-volume bars (Lopez de Prado, 2018) built from TBBO (trade-sampled top-of-book) events.

2 Data

The panel consists of CME-cleared TBBO data for the front-month Ether futures contract (CME product code ETH), covering February 2021 to August 2026. Bars never span a contract roll, and the mid-quote change across each roll is excluded from the regression.

Bars are formed on a one-minute grid: TBBO events are aggregated to minutes, and a bar closes at the first minute at which its cumulative dollar volume crosses a threshold set to one

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tenth of the median in-sample daily dollar volume, a target of ten bars per day, and held fixed out of sample.

For each bar the following quantities are computed:

- **ofi**: signed level-1 order flow imbalance, following the construction of Cont et al. (2014). For each TBBO update i , let p_i^B, p_i^A denote the best-bid and best-ask quotes and q_i^B, q_i^A their sizes. The per-event OFI contribution is $e_i = e_i^B + e_i^A$, where

$$e_i^B = \begin{cases} +q_i^B & \text{if } p_i^B > p_{i-1}^B \\ -q_{i-1}^B & \text{if } p_i^B < p_{i-1}^B \\ q_i^B - q_{i-1}^B & \text{if } p_i^B = p_{i-1}^B \end{cases}, \quad e_i^A = \begin{cases} -q_i^A & \text{if } p_i^A < p_{i-1}^A \\ +q_{i-1}^A & \text{if } p_i^A > p_{i-1}^A \\ -(q_i^A - q_{i-1}^A) & \text{if } p_i^A = p_{i-1}^A. \end{cases}$$

The bar OFI is the unnormalized sum over all events in the bar,

$$O_t = \sum_{i \in \text{bar } t} e_i.$$

- **mid.close**: the mid quote $(p^B + p^A)/2$ at the bar boundary.
- **d.mid**: the change in mid quote since the previous bar.

3 Price-impact regression

Following Cont et al. (2014) and Silantyev (2019), bar mid-quote change is regressed on contemporaneous and one-bar lagged OFI:

$$\Delta p_t = \alpha + \beta_0 O_t + \beta_1 O_{t-1} + \varepsilon_t, \quad (1)$$

where $\Delta p_t = m_t - m_{t-1}$ is the change in the mid quote between bar boundaries and O_t is the bar OFI, so β_0 is in dollars of mid-quote change per contract of net imbalance. Standard errors are Newey-West HAC (Newey and West, 1987) with five lags. Table 1 reports the estimates on the full panel, the 2021-02 to 2023-12 in-sample window, and the 2024-01 to 2026-08 out-of-sample window.

Window	n (bars)	$\hat{\beta}_0$	HAC $t(\beta_0)$	$\hat{\beta}_1$	HAC $t(\beta_1)$	R^2
Full sample	41,055	+0.247	+70.2	-0.027	-14.4	0.29
In-sample 2021–23	12,344	+0.300	+44.9	-0.025	-7.0	0.31
Out-of-sample 2024–26	28,711	+0.215	+58.1	-0.026	-13.3	0.30

Table 1: OLS, Newey-West HAC (five lags). $p < 10^{-6}$ on both coefficients in every window.

This regression is contemporaneous: it measures how the mid quote moves together with the flow inside a bar (median out-of-sample bar duration ≈ 17 minutes), not whether flow forecasts the next bar. Regressing the next bar’s mid-quote change on lagged OFI alone gives $R^2 < 0.001$ out of sample. Whether the flow carries predictive content at horizons inside the bar is a question the bar regression cannot resolve.

Figure 1 shows the out-of-sample relation. The in-sample and out-of-sample R^2 are nearly identical (0.31 and 0.30), while the slope itself is less stable, moving from +0.300 in sample to +0.215 out of sample. Both R^2 figures are below the two-thirds level of Cont et al. (2014) for liquid US equities, albeit at a different sampling frequency; Section 4 repeats the comparison at their ten-second interval.

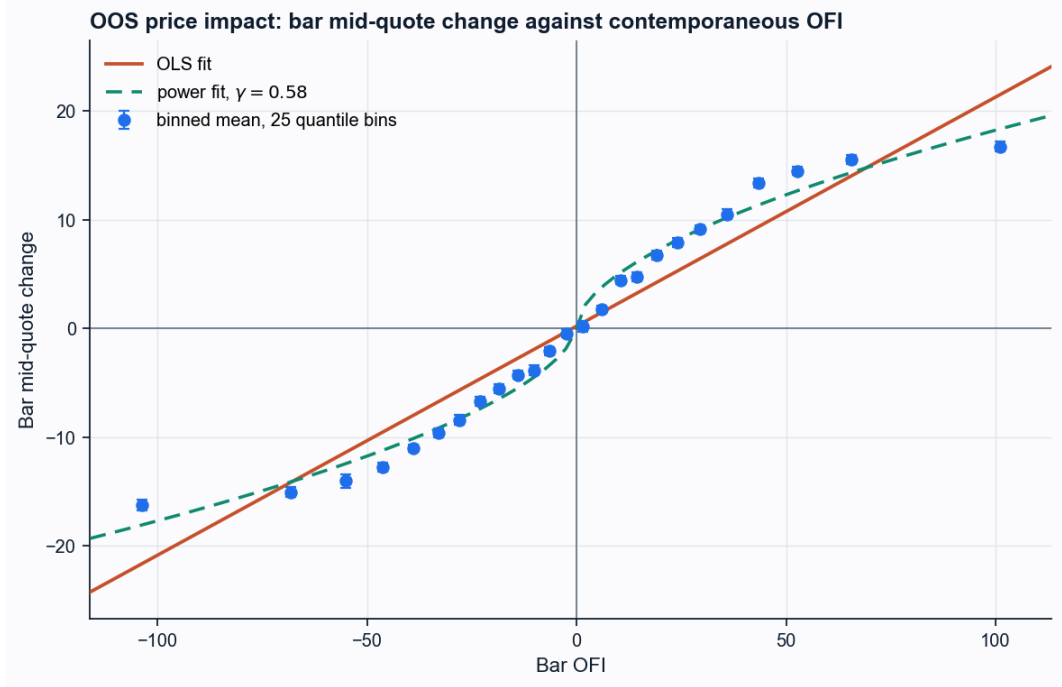


Figure 1: Out-of-sample bar mid-quote change against contemporaneous OFI, 25 quantile bins with ± 1 standard error, the least-squares fit of Δp_t on O_t alone, and the power fit with the exponent $\gamma = 0.58$ of Section 5. The relationship is monotonic, with impact per unit of flow attenuating in the tails: both extreme bins lie below the straight fitted line in absolute value, and the power fit tracks them. Full estimates are in Table 1.

4 Robustness to the bar scale

The ten-bars-per-day calibration is a convention rather than a fitted parameter, and the explanatory power does not depend on it. Rebuilding the panel at eighteen targets from one to one thousand bars per day, with the threshold recomputed on the in-sample window at each scale, leaves the out-of-sample R^2 between 0.29 and 0.32 everywhere, while $\hat{\beta}_0$ rises from +0.17 on the coarsest bars to +0.25 on the finest (Figure 2).

Median durations run from almost four hours down to the panel’s one-minute resolution, which binds beyond roughly two hundred bars per day: past that point most bars complete within a single minute, so the finest scales are effectively one-minute time bars.

Because the one-minute floor is a property of the panel, not the data, the sweep is repeated on a ten-second event grid rebuilt from the raw TBBO stream for the out-of-sample window; the rebuilt buckets aggregate back to the minute panel exactly. Dollar-volume bars on this grid, from ten to five thousand bars per day, hold the out-of-sample R^2 between 0.31 and 0.32 (Figure 2).

On plain ten-second intervals, the sampling grid of Cont et al. (2014), the regression gives $\hat{\beta}_0 = +0.26$ ($t = 189$, $R^2 = 0.32$) on the 1.6 million ten-second intervals that contain at least one trade. The gap to the equity two-thirds is therefore not an artifact of bar size, though it is partly one of estimation design: their two-thirds is a mean over half-hour subsample regressions, while the 0.32 pools the full out-of-sample window under one coefficient.

Estimated their way, the mean R^2 over half-hour windows on the ten-second grid is 0.54 (windows with at least thirty such intervals), with the caveat that TBBO observes the book only at trade instants. The interquartile range across windows is 0.48 to 0.62, and the yearly means are 0.53 to 0.55.

Beyond about five bars per day the lagged coefficient shrinks steadily as bars get finer, and on plain ten-second intervals it is indistinguishable from zero ($t = -0.3$). The small negative $\hat{\beta}_1$ of Table 1 thus reflects netting within coarse bars rather than a persistent reversal in the flow.

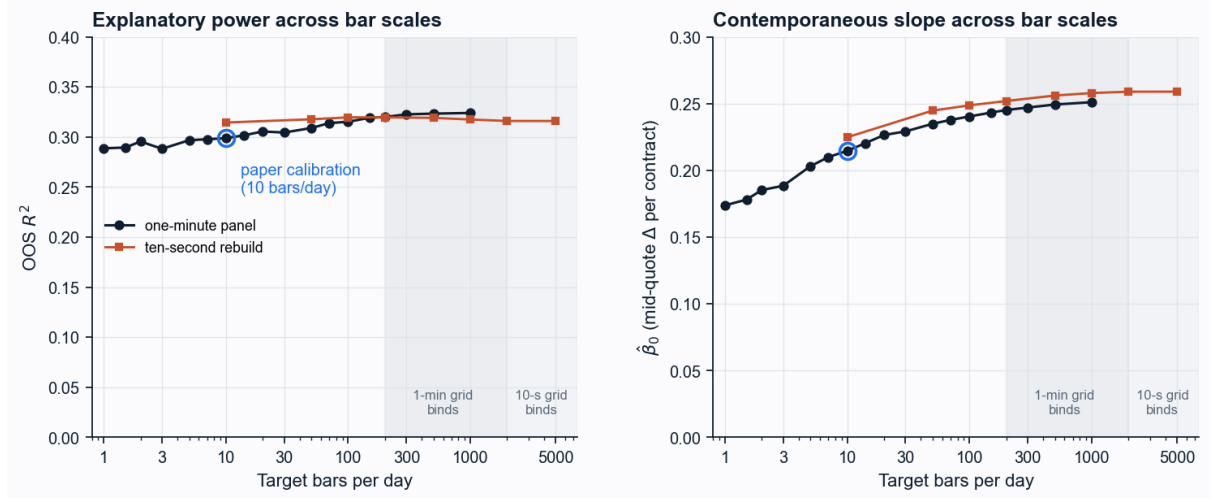


Figure 2: Out-of-sample R^2 (left) and contemporaneous slope $\hat{\beta}_0$ (right) as the bar threshold is varied, recalibrated on the in-sample window at each scale: one to one thousand target bars per day on the one-minute panel, and ten to five thousand on the ten-second rebuild. The circled point is the ten-bars-per-day calibration used throughout the paper; in the shaded bands each grid’s resolution binds: past 200 bars per day most minute-panel bars complete within a single minute, and past 2000 on the ten-second rebuild about half the bars collapse to single buckets.

5 Shape and state dependence

The tail attenuation in Figure 1 can be given a number. Replacing the linear term with a power law,

$$\Delta p_t = \alpha + b \text{sign}(O_t) |O_t|^\gamma + \varepsilon_t, \quad (2)$$

and profiling γ by least squares gives $\hat{\gamma} = 0.58$ (day-block bootstrap 95% confidence interval 0.54 to 0.60), and raises the univariate out-of-sample R^2 from 0.29 to 0.32.

Impact in this market is concave in flow, with an exponent well below one and consistent with the $3/5$ that Almgren et al. (2005) estimate for equity metaorder impact, though their object is a parent order worked over time and ours is the net flow of a bar. The linear $\hat{\beta}_0$ of Table 1 is the local slope of this curve at typical flows. The relation is symmetric in sign: separate slopes on positive and negative OFI come out at +0.215 and +0.214, and their difference carries $t = 0.06$.

Impact also moves with the state of the book. Splitting the out-of-sample bars into terciles by the quoted spread at the bar boundary and estimating the regression in basis-point returns, $\hat{\beta}_0$ rises monotonically from 0.58 bp per contract in the narrow-spread tercile (median 2.2 bp) through 0.76 in the middle to 1.03 in the wide (median 5.9 bp). A continuous interaction of OFI with the spread carries $t = 17$ and holds within each calendar year separately (t of 8.3, 8.2, and 7.0 in 2024, 2025, and 2026).

Wide spreads and large moves are both symptoms of volatile periods, so the interaction is re-estimated against that confound: with the spread lagged one bar it carries $t = 10.5$, with a control for trailing twenty-bar realized volatility added it carries $t = 7.1$, and it holds inside each volatility tercile (t of 2.8, 4.5, and 5.8 from low to high volatility). A wide spread marks a thin book, so this is the inverse-depth scaling of Cont et al. (2014), observed at the touch.

6 Conclusion

The contemporaneous price impact of OFI is strong on CME Ether futures, its explanatory power carries out of sample essentially unchanged, and the reading is the same at every scale the data support, from ten-second intervals to bars spanning hours. The estimates describe how this market absorbs order flow.

TBBO is trade-sampled, so the book is observed only at trade instants, and it carries no depth beyond the touch. The ten-second grid already measures inside the bar and finds no predictive content; a quote-level feed would show whether that null holds between trades, and its depth would permit a proper treatment of execution. These are the natural extensions.

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